

Optimizing Score Thresholds in Score-explained Heterogenous Treatment Effect Models



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Problem Setting

We consider a setting where treatment assignment is determined by some type of score Q , where only people above (or below) a given threshold ϕ are assigned treatment, with heterogenous effect W . We also consider covariates X .

$$Y = W\mathbf{1}(Q > \phi) + \theta^\top X + \epsilon$$

Probation Dataset

An example of this setting is probation, where students are put on probation if their first GPA is below some threshold, with the goal of an improved second year. We use data from a large Canadian university.

In our model, Q is first year GPA (minus probation threshold), Y is the second year GPA, and X includes variables such as gender, age, and high school grades.

Our goal is to optimize this GPA threshold for probation.

Standard Approach

Q: How do we measure the treatment effect?

We can not compare the treatment group with the control group directly! A standard approach is to **consider a small neighborhood around the threshold**, which has drawbacks:

1. This method only uses data from a small neighborhood of the threshold, so estimates for the treatment effect do not converge at a $\sqrt{n}$ rate.
2. This method only estimates the treatment effect at the current threshold and not at other scores, so it can not make a prescriptive claim for an optimal threshold.

Recent Work

Previous work [1,2] considered modeling $Q = \gamma^\top X + \eta$ where the error η represents some form of latent individual “ability.” They then use this to estimate the treatment effect (constant or homogenous) in a $\sqrt{n}$ rate.

Methodology

1. Define $\alpha(\eta) = E[W|\eta]$ and $\beta(\eta) = E[\epsilon|\eta]$.
2. Using the same linear model for Q , we estimate the natural ability $\hat{\eta}$.
3. In the treatment group, we have partial linear model

$$Y = \alpha(\eta) + \theta^\top X + \beta(\eta) + R_1$$

and in the control, we have

$$Y = \theta^\top X + \beta(\eta) + R_2,$$

with mean 0 errors R_1, R_2 . Thus, we can estimate α by estimating $\alpha + \beta$ and β nonparametrically and taking the difference.

4. We then optimize a social utility function

$$U(\phi) = E[(W - c)\mathbf{1}(Q > \phi)]$$

for some given cost c .

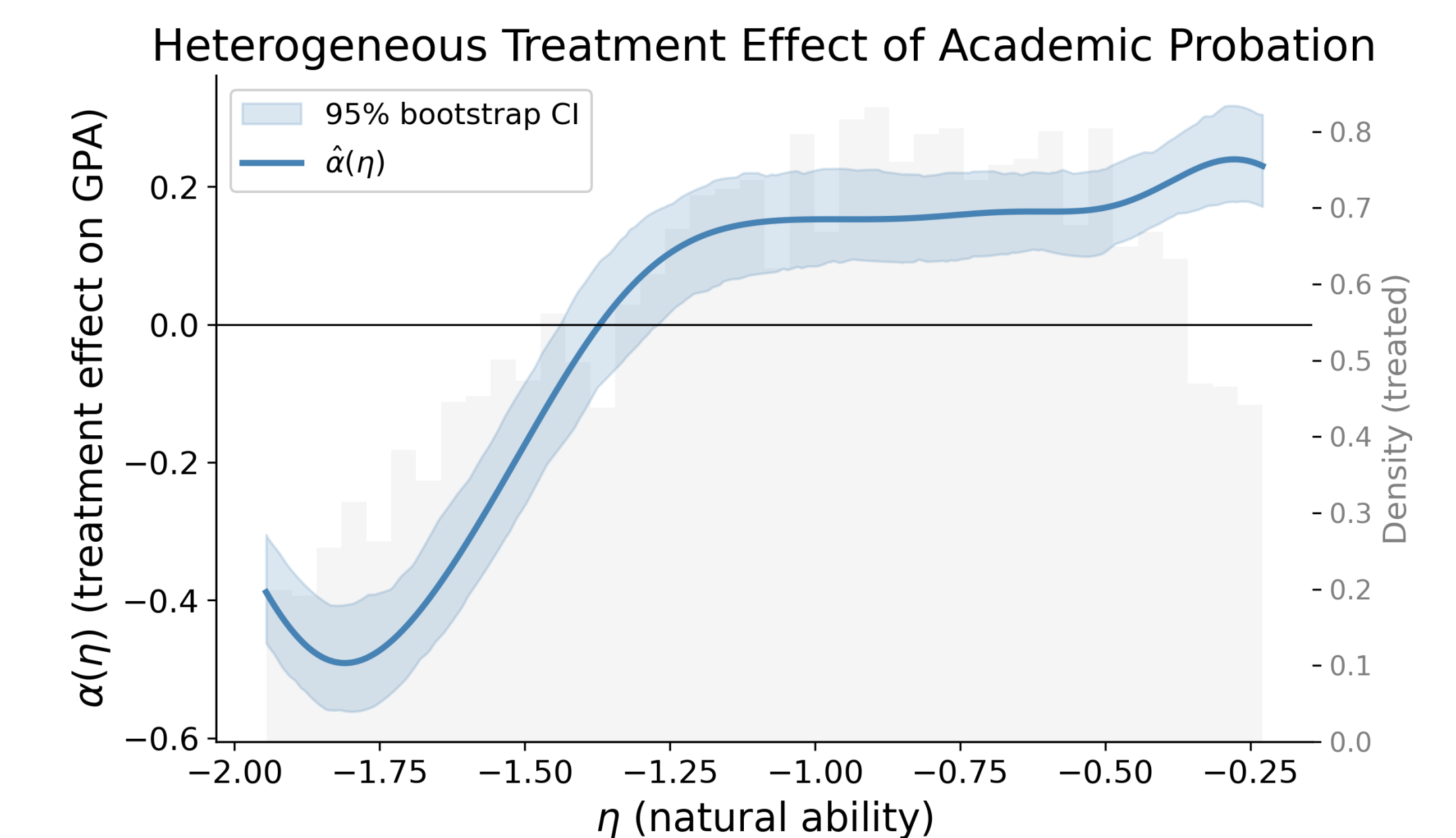
Our Contribution

- Without any parametric assumptions on α and β , we are able to estimate the treatment effect function $\alpha(\eta)$.
- With the above algorithm, we can estimate the optimal threshold ϕ^* at a $\sqrt{n}$ rate with undersmoothing.

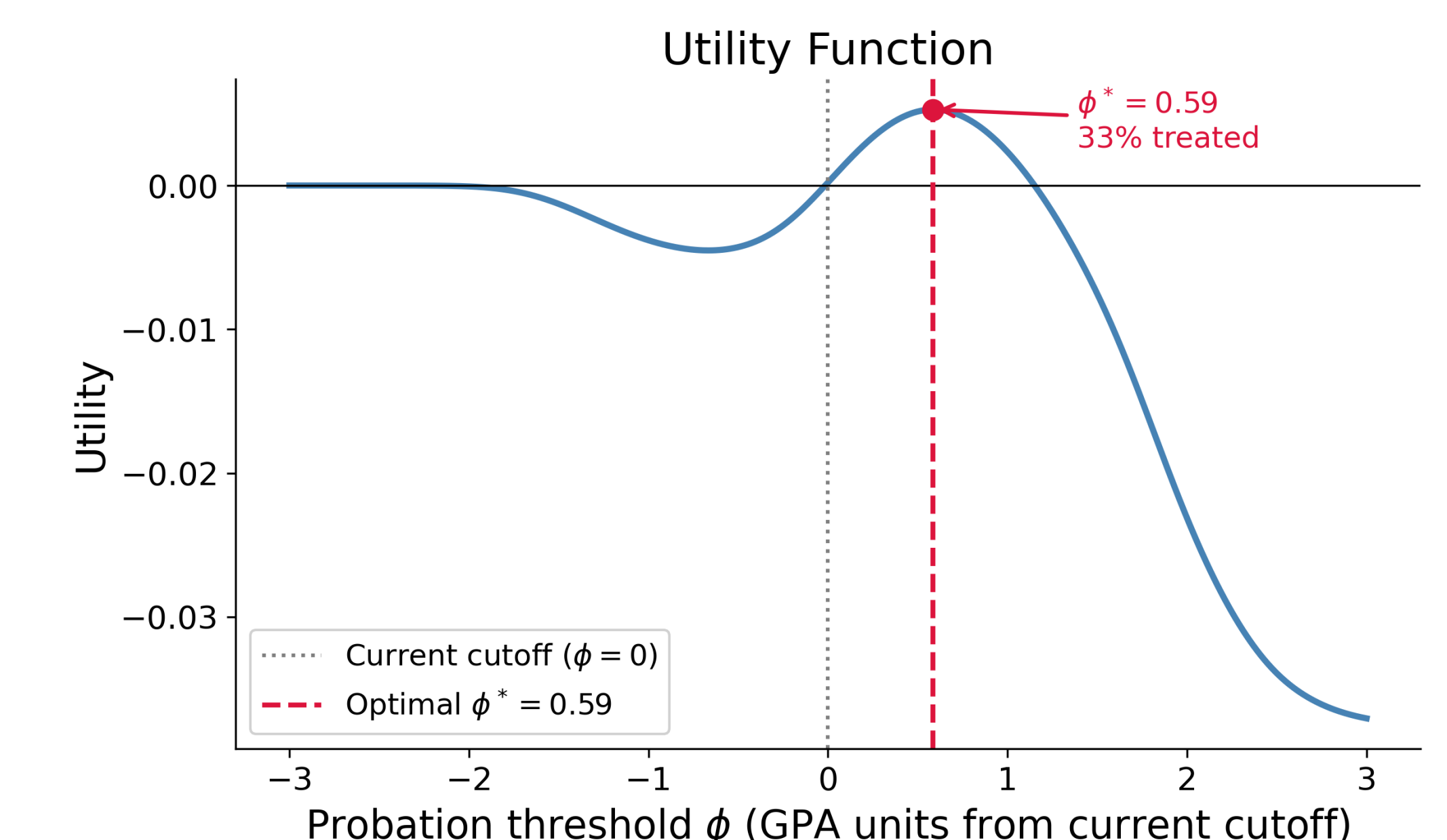
Experimental Results

Probation carries a cost. First, it increases the rate of dropping out, which we consider as a 0 second year GPA. We also consider giving higher GPA students probation more costly than a lower GPA student, so we impose cost $c = \frac{GPA}{25}$.

We estimate the treatment effect function α with bootstrap confidence bounds.



We then optimize the GPA threshold, suggesting that utility is maximized if the GPA threshold is increased by 0.59.



References

- [1] Moulinath Banerjee Debarghya Mukherjee and Ya'acov Ritov. Estimation of a score-explained non-randomized treatment effect in fixed and high dimensions, 2021.
- [2] M. Banerjee K. Wibisono, D. Mukherjee and Y. Ritov. Estimation and inference for the average treatment effect in a score-explained heterogeneous treatment effect model, 2025.